Retrieval, Attention, and LLM

Organization of Web Information

by

Aniket Waghela 25M2104
Ninad Parikh 25M2103
Riddhankur Sarkar 25M2106

Department of Computer Science and Engineering
Indian Institute of Technology, Bombay
Mumbai 400076 (India)

April 5, 2026

Contents

1 Introduction **2**

2 Classical Retrieval **3**
2.1 Methodology . . . 3
2.2 Results . . . 3
2.3 Discussion . . . 3

3 Attention-Based Retrieval and Positional Effects **4**
3.1 Methodology . . . 4
3.2 Retrieval Results . . . 4
3.3 Gold Tool Position vs. Attention Score Analysis . . . 4

4 Retrieval Heads **6**
4.1 Phase 1: Head Selection Strategy . . . 6
4.2 Phase 2: Retrieval Using Selected Heads . . . 6
4.3 Comparison Across Parts 1, 2, and 3 . . . 7
4.4 Analysis and Implications . . . 7

5 Bonus: Effect of Varying the Number of Selected Heads **8**
5.1 Motivation . . . 8
5.2 Selected Heads for Each K . . . 8
5.3 Results . . . 8
5.4 Discussion . . . 8

6 Bonus: Alternative Head Selection Strategies **10**
6.1 Motivation . . . 10
6.2 Description of Approaches . . . 10
6.3 Selected Heads per Approach . . . 11
6.4 Results . . . 11
6.5 Discussion . . . 11

7 Conclusion **13**

Github Link : `https://github.com/Ninad1306/Retrieval-Attention-and-LLM`

Chapter 1

Introduction

This assignment explores how language models can be used to select the correct tool given a natural language query. For each query, we are given a gold tool, each with a description, and our goal is to retrieve the correct tool corresponding to the query.

The assignment is divided into three parts plus two bonus tasks:

- **Part 1:** Classical retrieval methods (BM25, MiniLM, UAE-large-v1)
- **Part 2:** Attention-based retrieval and positional effects using LLaMA-3.2-1B-Instruct
- **Part 3:** Identifying and using retrieval-relevant attention heads
- **Bonus 1:** Effect of varying the number of selected heads ($K = 10, 20, 30$)
- **Bonus 2:** Alternative head selection strategies (normalized attention, rank-based, reciprocal rank)

Chapter 2

Classical Retrieval

2.1 Methodology

In Part 1, we implement and evaluate three classical retrieval methods, all following the standard encode-independently-then-compare paradigm:

1. **BM25** (Sparse Retrieval): A classical term-frequency based retrieval model that scores documents based on lexical overlap with the query. BM25 does not use dense representations; instead, it relies on exact term matching with TF-IDF-style weighting.
2. **msmarco-MiniLM** (Dense Retrieval): A compact, fine-tuned dense encoder based on the MiniLM architecture, trained on the MS MARCO dataset for passage retrieval. The query and each tool description are encoded independently into dense vectors, and cosine similarity is used to rank tools.
3. **UAE-large-v1** (Dense Retrieval): A large universal adaptive embedding model designed for general-purpose retrieval. Like MiniLM, it encodes queries and tools independently, but leverages a significantly larger and more expressive backbone.

2.2 Results

Table 2.1 reports Recall@1 and Recall@5 for all three baselines on the test query set.

Table 2.1: Classical Retrieval Results (Part 1)

Method	**Recall@1**	**Recall@5**
BM25	0.1882	0.3452
MiniLM	0.3882	0.6106
UAE-large-v1	0.6158	0.8690

2.3 Discussion

The results clearly demonstrate that dense retrieval substantially outperforms sparse BM25, particularly for this task where query and tool descriptions may use semantically related but lexically different vocabulary. UAE-large-v1 achieves the best performance, with a Recall@1 of 0.6158 and Recall@5 of 0.8690, indicating that the correct tool appears in the top 5 results nearly 87% of the time. MiniLM offers a strong middle ground, substantially better than BM25 while being computationally cheaper than UAE-large-v1.

Chapter 3

Attention-Based Retrieval and Positional Effects

3.1 Methodology

In Part 2, we move from classical retrieval to using the internal attention weights of an LLM (LLaMA-3.2-1B-Instruct) as a retrieval signal.

Additionally, we analyze the relationship between the position of the correct (gold) tool in the prompt and the attention score it receives, aggregating results across all test queries to reveal positional biases.

3.2 Retrieval Results

Table 3.1 shows the attention-based retrieval performance.

Table 3.1: Attention-Based Retrieval Results (Part 2)

Method	**Recall@1**	**Recall@5**
Attention (All Heads, All Layers)	0.0122	0.2758

The attention-based approach with all heads aggregated performs significantly below the classical retrieval baselines, suggesting that raw attention averages across all layers and heads are noisy retrieval signals. Not all attention heads contribute meaningful retrieval information; many heads attend to positional, syntactic, or other non-retrieval-relevant features.

3.3 Gold Tool Position vs. Attention Score Analysis

Figure 3.1 shows how the average attention score assigned to the correct (gold) tool varies as a function of its position in the prompt.

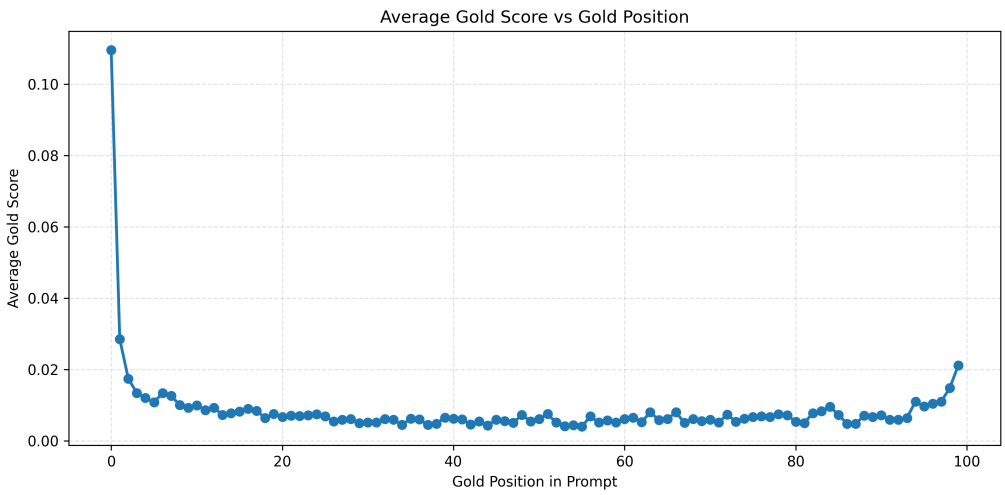

Figure 3.1: Average Gold Score vs. Gold Position in Prompt (Part 2). The x-axis represents the position of the correct tool (0-indexed), and the y-axis represents the average attention score it receives.

The plot reveals a clear **primacy bias**: the attention score assigned to the gold tool is dramatically higher when the tool appears at position 0 (the very beginning of the prompt) compared to all other positions. There is also a smaller **recency effect**: gold tools placed at the very end of the prompt (positions 95–100) also receive slightly elevated attention scores. Between these two extremes, attention to the gold tool is relatively flat and uniformly low. This U-shaped (or "Lost-in-the-Middle") pattern is consistent with the well-known lost-in-the-middle phenomenon in LLMs: information at the beginning and end of long contexts is attended to more than information in the middle.

Chapter 4

Retrieval Heads

4.1 Phase 1: Head Selection Strategy

In Part 3, we investigate whether a small subset of attention heads can act as effective retrievers. The hypothesis is that certain heads specialize in semantic matching between query and document, and filtering to only these heads can improve retrieval quality.

Head Selection Procedure:

We use the training query set (200 queries) to identify the most retrieval-relevant heads. For each query, we run the model and extract attention matrices for all layers and heads. We score each head by the raw sum of attention mass it directs from query tokens toward the gold tool's token span, accumulated across all training queries. Heads that consistently assign high total attention to the correct tool receive high scores. After ranking all heads by this per-head retrieval score, we select the top $K = 20$ heads.

Selected Heads (K=20):

The 20 selected retrieval heads are listed below as (layer_id, head_id) pairs:

(7, 12)	(10, 9)	(10, 26)	(1, 23)
(8, 19)	(6, 11)	(8, 13)	(8, 20)
(12, 13)	(10, 1)	(10, 13)	(12, 15)
(10, 23)	(12, 2)	(8, 31)	(6, 8)
(5, 11)	(11, 27)	(10, 14)	(10, 20)

The selected heads are concentrated in layers 8, 10, and 12, with a few in layers 5–7. This is consistent with the general finding that middle-to-late transformer layers develop more task-relevant semantic representations.

4.2 Phase 2: Retrieval Using Selected Heads

Using only the 20 selected heads, we repeat the attention-based scoring procedure from Part 2. The score for each tool is computed by averaging attention weights from query tokens to tool tokens, but only across the selected 20 heads. Table 4.1 reports Recall@1 and Recall@5 for selected heads.

Table 4.1: Retrieval Using Selected Heads (Part 3, K=20)

Method	**Recall@1**	**Recall@5**
Attention (Selected 20 Heads)	0.1082	0.5428

The selected-head approach substantially outperforms the all-heads baseline from Part 2 (Recall@1: 0.1082 vs. 0.0122; Recall@5: 0.5428 vs. 0.2758), demonstrating that retrieval-relevant heads encode meaningfully different information than the average over all heads.

4.3 Comparison Across Parts 1, 2, and 3

Table 4.2 provides a comprehensive comparison of all methods.

Table 4.2: Comprehensive Comparison of All Retrieval Methods

Part	**Method**	**Recall@1**	**Recall@5**
Part 1	BM25	0.1882	0.3452
Part 1	MiniLM (Dense)	0.3882	0.6106
Part 1	UAE-large-v1 (Dense)	**0.6158**	**0.8690**
Part 2	Attention (All Heads)	0.0122	0.2758
Part 3	Attention (20 Selected Heads)	0.1082	0.5428

4.4 Analysis and Implications

Several important observations emerge from the comparison:

1. **Attention-based retrieval (all heads) is weaker than classical methods.** Using raw averages over all attention heads produces a noisy signal. Most heads encode information irrelevant to semantic matching (e.g., positional patterns, syntactic structure), diluting the signal from the few retrieval-relevant heads.

2. **Head selection substantially recovers performance.** Filtering to only 20 heads improves Recall@1 by nearly 9× (0.0122 → 0.1082) and Recall@5 by nearly 2× (0.2758 → 0.5428). This confirms the hypothesis that a sparse subset of heads specializes in retrieval-relevant attention.

3. **Classical dense retrieval still dominates for this task.** Even with head selection, the attention-based method (Recall@1: 0.1082) does not match even BM25 (0.1882), let alone MiniLM or UAE-large-v1. This suggests that joint-attention is not as effective as encode-then-compare for ranking without retrieval fine-tuning.

4. **The concentration of selected heads in layers 8–12** suggests that semantic retrieval-relevant information is encoded more in middle-to-late layers, aligning with the broader literature on probing transformer representations.

5. **The lost-in-the-middle effect** (observed in Part 2) implies that the model's attention mechanism is not position-invariant for long contexts, which fundamentally limits joint-prompt retrieval approaches.

Chapter 5

Bonus: Effect of Varying the Number of Selected Heads

5.1 Motivation

The main experiment used $K = 20$ selected heads. We investigate how the choice of K affects retrieval performance by testing $K \in \{10, 20, 30\}$, keeping all other settings identical (same head scoring criterion, same 200 training queries, same test set of 5000 queries).

5.2 Selected Heads for Each K

For $K = 10$, the selected heads are the top-10 from the same head ranking used for $K = 20$ (a strict subset):

K=10: (7,12), (10,9), (10,26), (1,23), (8,19), (6,11), (8,13), (8,20), (12,13), (10,1)

For $K = 30$, the 10 additional heads (ranks 21–30) that are appended beyond $K = 20$ are:

Additional (K=21 to 30): (10,15), (8,30), (11,7), (11,20), (1,7), (15,30), (13,2), (13,20), (4,16), (1,24)

5.3 Results

Table 5.1 shows Recall@1 and Recall@5 for each value of K.

Table 5.1: Bonus 1 — Effect of Number of Selected Heads K (Raw Attention Sum scoring)

K (num. heads)	Recall@1	Recall@5
10	0.0940	**0.6086**
20	**0.1082**	0.5428
30	0.0340	0.5162

5.4 Discussion

The results reveal an interesting non-monotonic pattern across K:

- $K = 20$ **gives the best Recall@1** (0.1082). It provides enough heads to capture diverse retrieval-relevant signals while avoiding the inclusion of noisy lower-ranked heads that hurt top-1 precision.
- $K = 10$ **gives the best Recall@5** (0.6086). The very top-10 heads carry the most concentrated and pure retrieval signal. With fewer but higher-quality heads, the model places the correct tool in the top 5 more reliably, even if it fails to rank it first consistently.
- $K = 30$ **degrades significantly on both metrics** (Recall@1: 0.0340, Recall@5: 0.5162). The extra 10 heads ranked 21–30 appear to be marginal or noisy heads whose inclusion corrupts the composite attention score and confuses the ranking.

This points to a clear **diminishing-returns** (and eventually negative-returns) phenomenon: the top handful of heads contain the most retrieval-relevant signal, and including more heads beyond a sweet spot introduces noise. There is also a Recall@1 vs. Recall@5 trade-off: fewer heads improve top-5 coverage, while a moderate number of heads better discriminates the single top-ranked tool. In practice, K should be tuned on a held-out validation set.

Chapter 6

Bonus: Alternative Head Selection Strategies

6.1 Motivation

The baseline head selection strategy (Approach 1) scores each head by the *raw sum* of attention it directs from query tokens to the gold tool across all training queries. This simple criterion has a key weakness: a head that gives high but *uniform* attention to all tools (including the gold one) will score highly, even though it provides no discriminative signal for ranking. We explore three alternative strategies that address this limitation.

6.2 Description of Approaches

Approach 1 — Raw Attention Sum (Baseline, Part 3): For each head (l, h) and each training query q, sum the total attention flowing from the query span to the gold tool's token span and accumulate across all queries:

$$\text{score}(l, h) = \sum_{q \in \mathcal{Q}_{\text{train}}} \sum_{i \in \text{query}_q} \sum_{j \in \text{gold}_q} \alpha_{l,h,i,j}$$

This does not penalize heads that attend broadly and uniformly to all tools.

Approach 2 — Normalized Gold Attention: We compute the *fraction* of attention the head gives to the gold tool relative to the total attention it gives to all tools combined. This penalizes uniform-attention heads and rewards selectively attending heads:

$$\text{score}(l, h) = \sum_{q \in \mathcal{Q}_{\text{train}}} \frac{\displaystyle\sum_{i \in \text{query}_q} \sum_{j \in \text{gold}_q} \alpha_{l,h,i,j}}{\displaystyle\sum_{d \in \text{all tools}} \sum_{i \in \text{query}_q} \sum_{j \in d} \alpha_{l,h,i,j} + \varepsilon}$$

Approach 3 — Rank-Based Scoring (Training Recall@1 per Head): For each training query, use each head independently to rank all tools by the attention it assigns them. Score the head by the number of training queries for which the gold tool is ranked at position 1 — directly optimizing the same metric used at test time:

$$\text{score}(l, h) = \sum_{q \in \mathcal{Q}_{\text{train}}} \mathbf{1}\big[\text{rank}_{l,h}(\text{gold}_q) = 1\big]$$

Approach 4 — Reciprocal Rank Scoring (Training MRR per Head): A softer version of Approach 3 that gives partial credit when the gold tool is ranked near the top but not first, using the Mean Reciprocal Rank (MRR) criterion:

$$\text{score}(l, h) = \sum_{q \in \mathcal{Q}_{\text{train}}} \frac{1}{\text{rank}_{l,h}(\text{gold}_q) + 1}$$

Heads that rank the gold tool at position 2 or 3 still receive meaningful credit, making the signal smoother than the binary Approach 3.

6.3 Selected Heads per Approach

All approaches use $K = 20$ heads and 200 training queries. Table 6.1 lists the 20 selected heads for each approach.

Table 6.1: Selected Heads (K=20) for Each Approach in Bonus 2

c

Approach	Selected Heads (layer, head)
1 (Baseline)	(7,12), (10,9), (10,26), (1,23), (8,19), (6,11), (8,13), (8,20), (12,13), (10,1), (10,13), (12,15), (10,23),
2 (Normalized)	(6,11), (10,1), (6,8), (5,11), (10,23), (12,13), (10,9), (6,9), (8,19), (12,15), (10,13), (8,20), (8,13), (10
3 (Rank-Based)	(6,11), (10,9), (10,1), (8,19), (4,16), (5,11), (2,20), (6,8), (6,9), (7,12), (10,23), (5,10), (12,13), (8,
4 (Recip. Rank)	(6,11), (7,12), (4,16), (8,19), (5,11), (10,9), (2,20), (10,1), (6,8), (10,23), (6,9), (5,10), (8,20), (8,1

Several heads appear consistently across *all four* approaches: (6,11), (10,9), (10,1), (8,19), (5,11), (8,13), (8,20), (12,13), and (7,12). These are robustly retrieval-relevant regardless of the scoring criterion. Approaches 3 and 4 additionally identify (4,16), (2,20), (6,9), (5,10), and (5,9) — heads missed by the raw-sum baseline — reflecting the discriminativeness captured by rank-based criteria.

6.4 Results

Table 6.2 reports Recall@1 and Recall@5 for all four approaches at $K = 20$.

Table 6.2: Bonus 2 — Recall Performance of Alternative Head Selection Strategies (K=20)

Approach	**Scoring Criterion**	**Recall@1**	**Recall@5**
1	Raw Attention Sum (Baseline)	0.1082	0.5428
2	Normalized Gold Attention	0.2268	0.5758
3	Rank-Based (Recall@1 on train)	**0.2790**	**0.5962**
4	Reciprocal Rank (MRR on train)	0.2634	0.5916

6.5 Discussion

The results reveal substantial and consistent improvements from switching head selection criterion:

- **Approach 2 (Normalized) more than doubles Recall@1** over the baseline (0.2268 vs. 0.1082). By normalizing gold attention by total tool attention, it filters out heads that attend broadly to all tools and selects only those that preferentially focus on the correct one. This is a simple but highly effective modification.

- **Approach 3 (Rank-Based) achieves the best Recall@1 (0.2790) and Recall@5 (0.5962)** among all head selection strategies. By directly optimizing per-head Recall@1 on training data, it selects heads that are most predictive of the test-time ranking objective. This can be viewed as a form of head-level supervision using training labels.
- **Approach 4 (Reciprocal Rank) closely follows Approach 3** (0.2634 Recall@1, 0.5916 Recall@5). The soft MRR criterion selects a nearly identical head set to Approach 3 (13 out of 20 heads overlap), suggesting both identify the same core retrieval heads. The small gap may be because the binary rank-1 criterion more precisely aligns with the Recall@1 test metric.
- **All three alternatives substantially outperform the raw-sum baseline**, confirming that *discriminativeness* (how much more a head attends to the gold tool compared to distractors) is the key factor in head selection, not raw attention magnitude.
- **Comparison with classical retrieval**: the best approach (Approach 3, Recall@1 = 0.2790) now surpasses BM25 (0.1882) and bridges a meaningful portion of the gap to MiniLM (0.3882). This demonstrates that with an appropriate selection criterion, LLM internal attention can be a competitive retrieval signal even without any fine-tuning.

Chapter 7

Conclusion

This assignment demonstrates the tradeoffs between classical and attention-based retrieval in the tool-selection setting. The key findings are:

1. **Classical dense retrieval (UAE-large-v1)** achieves the best overall performance, benefiting from dedicated retrieval training objectives.
2. **Naive all-head attention aggregation** is a very weak retrieval signal, heavily corrupted by syntactic and positional attention patterns and exhibiting a severe lost-in-the-middle positional bias.
3. **Head selection** is critical: filtering to a small subset of retrieval-relevant heads provides order-of-magnitude improvements over the all-head baseline. The optimal K lies around 10–20 for this model and task, with a clear degradation at $K = 30$.
4. **Head selection strategy matters greatly.** The simple raw-sum baseline is significantly outperformed by discriminativeness-aware criteria. Rank-based selection (directly optimizing per-head Recall@1 on training data) is the most effective approach, followed closely by reciprocal-rank and normalized attention scoring. The best strategy (Approach 3) pushes Recall@1 to 0.2790, surpassing BM25.
5. **A core set of retrieval heads** — particularly in layers 6–12, including heads such as (6,11), (10,9), (8,19), and (7,12) — is consistently identified across all selection strategies, suggesting these heads genuinely specialize in semantic query-document matching within LLaMA-3.2-1B-Instruct.